Syllabus

STAT 141: Introduction to Probability and Statistics
Reed College
Fall 2026

Professor Megan Ayers

Useful Information

Lecture and Lab Instructor: Megan Ayers (she/her)

- Email: meganayers@reed.edu
- Office hours: See Moodle for weekly office hour times and locations. I am also happy to meet ad hoc by appointment if you have conflicts during the scheduled times. Please email me or send me a Slack message to find a time to meet.

Course Assistants: We have course assistants (CAs) for STAT 141 this semester! CAs will each offer weekly office hours - these are a great space to meet to collaborate with your classmates and to get questions answered. Times and locations of course assistant office hours can be found on Moodle.

Links and course resources:

The course website, https://megan-k-ayers.github.io/stat141-fall26/, includes information on the course, lecture slides, public course materials, and links to all other course resources:

- Textbooks (links on this Syllabus page),
- RStudio Server, for working on homework and lab assignments (icon in upper right),
- Slack, for course correspondence (icon in upper right),
- Gradescope, for turning in assignments (icon in upper right), and
- occasionally, Moodle, for information about meetings and private course materials (icon in upper right).

Note that **the entire course website (including lecture slide material) is searchable!!** This may come in handy when completing assignments. Click on the magnifying glass icon in the upper right corner of the website to search.

Meeting Times: We'll have a lecture-style meeting three times a week, and a lab meeting once a week. Please see the course Moodle page or the Reed schedule of classes to verify the times and locations for your section.

Communication: I am most easily reached over email on weekdays between 9am and 5pm. I will do my best to answer your email within one school day. *For urgent matters, please email me rather than sending a Slack message.* For non-urgent course-related questions, particularly questions that fellow classmates or course assistants may know the answer to, your question may be answered more quickly in a public channel on Slack, which I will also monitor.

Learning Outcomes

In this course, you will learn how to think critically with data by engaging in the entire data analysis process through theory, computation, and practice. While most of our time will be spent learning techniques related to the *Exploration and Visualization* step and the *Modeling and Inference* step, you will also practice the other important components of analyzing data. Furthermore, since computation is an integral part of modern statistical work, you will learn to write `R` code – leveraging the `RStudio` user interface – to analyze data and will become familiar with several tidyverse `R` packages. You will also develop a reproducible and shareable workflow by using `Quarto` documents for all analyses.

By the end of the course, you will have improved your data acumen and your ability to think statistically. More concretely, you will be better able to accomplish the following tasks, which have been broken down by steps of the data analysis workflow:



Question formulation:

- Translate a research problem into a set of questions that can be answered with data.
- Formulate data questions as measurable statements about statistics and parameters.

Data acquisition:

- Determine the necessary data to conduct analyses.
- Reflect on how design structures and data collection impact potential conclusions.
- Identify potential ethical concerns surrounding data collection and data privacy.

Data wrangling:

- Explore datasets to determine what wrangling may be required (e.g., removing missing values, filtering out variables or observations, collapsing categories of a categorical variable).
- Apply basic data wrangling operations.

Exploration and Visualization:

- Understand key principles of designing and creating effective data visualizations.

- Create graphs from data, and appropriately draw conclusions from graphs.
- Compute and interpret summary statistics.

Modeling and Inference:

- Understand and be able to explain key probabilistic and inferential concepts, such as, sampling, variability, random variables, distributions, confidence, and significance.
- Use probability theory and data to recognize patterns and associations between variables.
- Create and assess the appropriateness of linear regression models for a given problem and set of data.
- Appropriately apply and draw inferences from a statistical analysis using both simulation-based and theory-based methods, including quantifying and interpreting the uncertainty in model estimates.
- Consider the ethical implications of various modeling practices.

Communicating Findings:

- Develop a reproducible workflow using Quarto documents.
- Interpret and communicate results of statistical analyses effectively for both a statistical and a non-statistical audience.
- Be able to reflect on the data involved in an analysis and show a curiosity for other ways of examining and thinking about the data.

Learning Materials & Tools

Textbooks: We'll use three textbooks for our course, all of which are freely available online:

- Statistical Inference via Data Science: A ModernDive into R and the tidyverse, Second Edition
  - Reading abbreviation: **MD**
- Introduction to Modern Statistics, Second Edition
  - Reading abbreviation: **IMS**
- OpenIntro Statistics, Fourth Edition
  - Reading abbreviation: **OI**

Technologies (R, RStudio, and Quarto): R is a free and open source programming language, RStudio is an Integrated Development Environment (IDE) which allows for streamlined use of the R programming language, and Quarto is a markdown language that allows for reproducible documents that include R code, text, images, and much more! We will access these technologies via the RStudio Server for this course. A laptop that can access the internet and use the RStudio Server is required for this course.

Please let me know ASAP if you do not have access to a personal computer!

Tutoring & Academic Support: I encourage you to reach out to Academic Support at Reed if you feel you're struggling or feeling overwhelmed with academics. They can help with study habits, note-taking, or time and stress management strategies. You may also want to take advantage of the free individual tutoring available by appointment. For more information, see https://www.reed.edu/academic_support/ or email `academic-support@reed.edu`.

Assignments, activities, and exams

We'll have a variety of assignments, activities, and assessments for this course. In particular:

- **Lab assignments**:
  - Every week we will have a lab assignment. The lab assignments will be worked on collaboratively during lab time on Thursdays, where you will be able to ask the instructor questions. They are designed to help you practice implementing concepts and tasks from lecture, particularly those that relate to coding in R. The following homework assignment will build on questions covered in lab.
  - Lab assignments will be made available for download from the course website before each lab.
  - You will submit the progress you made on each lab assignment via Gradescope at the end of your section's lab time. Labs will be assessed as part of your participation grade. You may receive some feedback on your lab work and will be shown correct solutions via Gradescope.
  - Late labs will not be accepted, and you need to be present in lab to receive credit for a lab submission. In the event of a missed lab, you should complete it outside of class to prepare for homework assignments. We'll be happy to discuss lab assignments in office hours.
- **Homework assignments**:
  - Every week we will have a homework assignment. Homework assignments will be posted on the course website on Thursday mornings.
  - These will be turned in via Gradescope by **9:00am on Thursay of the following week**.
  - Homework assignments will largely build on or continue content from that week's lab, and may also cover additional material from lecture.
  - You will receive feedback on your homework assignments, and scores will be assigned based on **completion** and **effort**.
  - Homework assignments represent a higher proportion of your grade than labs, but a relatively small proportion overall.

  - Engaging with homework assignments and discussing them in office hours will be the best way to study for the next week's quiz!
  - Grades and feedback are given based on your submitted PDF, and regrade requests will not be accepted. **Be sure to check that your rendered PDF legibly displays all of your work before submitting homework assignments to avoid formatting issue frustration!**

- **Quizzes**:
  - Almost every week at the beginning of lab we will have a short quiz, based closely on the material from the most recent homework assignment (ex. Quiz 1 will be based on content from HW 2).
  - Cumulatively, quizzes have the largest impact on your grade. However, there will be 10 quizzes throughout the semester, so each individual quiz will be low-stakes.
  - Your lowest quiz grade will be dropped - this may come in handy in the event of an expected or unexpected absence, as quizzes cannot be made up.
  - Quizzes will be given in written format and will be entirely closed-notes.
  - While you will not be asked to use a computer to write code in R during quizzes, you may be asked to write code by hand, describe coding steps qualitatively, or explain code that is shown to you.
  - You will have the opportunity to submit quiz revisions within 1 week of feedback being posted to regain partial credit.

- **In-class activities**:
  - Lectures and labs will often involve some in-class activities. Including but not limited to:
    * group activities,
    * independent activities,
    * participation-based learning assessments.

- **Readings**:
  - Readings will be posted ahead of every lecture. These may include a mix of content from the course textbooks and content from past lectures to review.
  - Readings are highly encouraged for deep engagement with course material and commitment of concepts to long-term memory.
  - The content in readings may seem repetitive with lecture content - this is by design! The more frequently you encounter and think through concepts that are presented in different ways, the better you will remember them, and the more deeply you will be able to engage with future material in the course that builds on them.

- **Final Exam**:
  - We'll have one exam (a final exam) for this course. The exam will consist of an in-person, cumulative, written exam.

- The final exam period will be three hours, and will be held during finals week (the official finals schedule is TBD).
- If you perform better on the final exam than cumulatively on quizzes, the exam will count more heavily towards your final grade. If you perform better across quizzes than on the final exam, the quizzes will count more heavily towards your final grade. In other words, whichever in-person assessment you perform best on will have the largest impact on your grade.
- No late exams will be accepted.

Attendance

Being actively present in lectures and labs is very important in this course. This requires attendance, and can look like:

- Active listening
- Contributing to small group or classroom level discussions
- Contributing to discussions on Slack.

Being consistently absent, late to class, or being consistently distracted during class (ex. using cell phone or doing homework during lecture), or consistently not participating in group work will detract from your participation grade.

That said, I understand that unexpected things happen, and that you may have an outside obligation come up during class time. To help with circumstances that might lead to an absence (expected or unexpected), everyone has 4 "free" lecture absences and 1 "free" lab absence (if you miss a lab where there is a quiz you will receive a 0 for it, but the lowest quiz score is dropped). You may use these free absences as needed, no questions asked. While I am here to listen, don't feel that you have to share anything personal to justify your absence.

Distribution Requirements

This course can be used towards your **Group III, "Natural, Mathematical, and Psychological Science"** requirement. It accomplishes the following learning goals for the group:

1. Use and evaluate quantitative data or modeling, or use logical/mathematical reasoning to evaluate, test or prove statements.
2. Given a problem or question, formulate a hypothesis or conjecture, and design an experiment, collect data or use mathematical reasoning to test or validate it.
3. Collect, analyze, and interpret data.

This course does not satisfy the "primary data collection and analysis" requirement.

Course Climate

We expect everyone in this class to strive to foster a learning environment that is equitable, inclusive, and welcoming. If you experience any barriers to learning, please come to Professor Megan Ayers or a college administrator with your concerns.

Code of Conduct:

We expect all members of Stat 141 to make participation a harassment-free experience for everyone, regardless of age, body size, visible or invisible disability, ethnicity, sex characteristics, gender identity and expression, level of experience, education, socio-economic status, nationality, personal appearance, race, religion, or sexual identity and orientation.

We expect everyone to act and interact in ways that contribute to an open, welcoming, inclusive, and healthy community of learners. You can contribute to a positive learning environment by demonstrating empathy and kindness, being respectful of differing viewpoints and experiences, and giving and gracefully accepting constructive feedback.[1]

Accommodations

Students come to this class with diverse needs and abilities. My goal is for this class to be accessible to all enrolled students. If you are planning to or are considering requesting a disability accommodation, I encourage you to contact Disability & Accessibility Resources (DAR), dar@reed.edu or 503-517-4917 as early as possible in the semester. Please be aware that requests may take several weeks to be processed through DAR. Once I receive accommodations requests via DAR I will reach out to discuss arrangements for each accommodation, but feel free to reach out to me to discuss accommodations at any time before or after this.

Policies

Technology policy:

- In STAT 141, we value a focused, distraction-free environment. Personal devices (including phones and headphones) and computers are generally not allowed in lecture to minimize digital distractions. However, technology is welcome for students who require digital tools for accessibility. We ask that these students limit their device use to essential in-class activities, such as note-taking or viewing digital readings. Non-academic use—including social media, messaging, and general web browsing—is not allowed, to ensure we remain fully engaged with the course material. Note that all lecture slides (including a condensed version for more efficient printing) will be made available ahead of lecture on the course website.

[1] This Code of Conduct is adapted from the Contributor Covenant, version 2.0.

- However, for some parts of the course, we will ask you to use your phone/tablet/computer to engage with course material (ex. you'll be asked to take attendance via QR code at the start of lecture, and to use your laptop to work on lab assignments and occasionally during lecture).
- Advice for note taking during lecture:
  - Rather than focusing on transcribing everything that is displayed or said, focus on summarizing content and ideas in your own words, recording any questions you have, and emphasizing any insights that helped you make sense of a concept.
  - Reference the slide number when taking each note so that you can contextualize it with the slides when reviewing your notes. It may be particularly helpful to make list of slides where code examples are provided for different tasks.

Late work policy:

- **Quizzes**: Since quizzes are taken in-person during lab, a missed quiz will be assigned a score of 0. Makeup quizzes will not be given, but, to help with various circumstances (expected and unexpected), the lowest quiz score will be dropped.
- **Homeworks**: To help with various circumstances (expected and unexpected), you have up to 3 additional homework extension days that you can use as you need, no questions asked. You are free to distribute the extension days to different assignments as you wish (e.g., 1 additional day for 3 HWs, 3 additional days for 1 HW) but the extension days must be rounded up to the nearest day (e.g., 2 extra hours = 1 extension day). When I see a late assignment submitted, I'll assume you intend to use extension days for it - you don't need to email me. If you've used all extension days, further late homework assignments will be accepted for up to 50% partial credit at any time before the end of the last day of class, Wednesday December 9th at 11:59pm. Late submissions of homework assignments are not guaranteed timely or detailed feedback - you are encouraged to meet with Megan or course assistants in office hours to clarify any questions or discuss solutions from any late homework assignments. ***Note:*** *Extension days only get "used up" if applied to an assignment submitted within the available window. For example, if you have one extension day remaining and submit an assignment two days late, that assignment will be subject to the late penalty, but you will still have your one extension day to use on a future assignment.*
- **Labs**: Lab assignments will be due at the end of each lab section and will count towards a small participation grade. If we ran out of time in lab to attempt all questions you don't need to rush to finish the lab before submitting it: just submit the progress that you made. You will have a grace period until the end of the day to submit your lab in the event of a PDF rendering error - please send me a quick email at the end of lab if you need to take this time to fix an error. In case of a lab absence, you are encouraged to prepare for homework assignments by completing labs on your own, but late submissions

of lab assignments will not be accepted. You may miss one lab without impact to your grade.

- **Exam**: The final exam cannot be made up.

Collaboration Policy and Academic Honesty:

Working with your classmates on difficult and interesting problems can not only help your learning, but also help you get to know each other! Therefore, I *highly* encourage you to collaborate on assignments. But, every piece of work you do *must* be your own. Copying other people's work or code is not acceptable (whether they are in our course or not), nor is allowing other people to copy your own work (whether they are in our course or not). The Honor Principle must guide your conduct in this class.

If you choose to collaborate with a classmate, **you must add their name to the top of your assignment**, and list them as a collaborator, e.g.:

Collaborator(s): Michael Pearce, Lenny Wainstein, Grayson White

But what is collaboration?:

For STAT 141, collaboration can look like: working with classmates together on a given problem, doing scratch work, helping each other get un-stuck on a part of a problem, and even coming to a solution together. However, you **must** write up your own problem solutions individually and cannot copy other's solutions (even those who you have collaborated with). You are expected to be able to explain the entirety your submitted solutions. When helping others with their assignment, you are encouraged to help them reason through concepts and to explain connections to lecture material and past assignments. However, you are expected to never explicitly share your completed solutions. You are expected to never distribute solutions obtained directly or indirectly via written feedback or office hours, including to students in this course or in other sections of 141. Copying solutions or code from **any source**, including a collaborator, classmate, an internet resource, or a generative AI tool (see the following section) is strictly prohibited.

AI and Internet Use Policy:

Artificial intelligence (AI) tools, such as ChatGPT, Claude, Co-Pilot, Gemini, and others are being used to generate code, analyze data, and much more. However, learning to think critically about a problem at hand, building resilience in moments of struggle and confusion, and engaging with your peers, tutors, and instructors when not understanding a concept or question are integral components of a liberal arts education. Further, a key goal of this course is for you to learn how to thoughtfully, ethically, and **independently** extract knowledge from data and engage in statistical reasoning. Therefore, the use of generative AI tools, such as ChatGPT and others, are strictly prohibited in any stage of the work process for this course.

The Honor Principle must guide your conduct in following this policy. If you have questions about whether a tool is allowed for this course, ask before using it.

While there are many ways to complete tasks in R, and many helpful internet resources for learning about them, this course focuses on a curated selection and is designed to be self contained. You may only use functions and packages that have been covered in lecture or lab on your assignments. You may use official online documentation resources (ex. https://tidyverse.org/packages/) to refresh yourself or learn more about functions that have been introduced in class. If you have questions at any point about other functions or packages, I (Megan) am happy to discuss! If you are unsure of how to approach a particular exercise or of what function to use, help is available to you! Please visit office hours, collaborate with your peers, or post in the Slack channel.

Acknowledgements

Thank you to Grayson White for allowing me to largely reuse the website structure he developed for Stat 141, and many of the related course materials (which in turn were influenced by the work of many others!). Thank you to Michael Pearce and Lenny Wainstein, many of whose previous Stat 141 course materials have also been adapted for this course.